

Paper planes to help fight forest fires

Dr Paul Pounds and a team of roboticists at the University of Queensland have designed two disposable, self-steering drones with sensor modules that can be dropped over forested areas to help monitor forest fires. One, called the Polyplane, looks like a paper plane, while the other, the Samara, mimics the maple seed. Both are made of biodegradable cellulose material. Both drones are self-deploying sensor modules that can relay back environmental conditions in difficult terrain. Dozens of them can be dropped across forested areas, to report back on atmospheric conditions that could indicate the start or movements of a bushfire—one of the biggest dangers lurking in Australian forests.

What's inside: Both drones are made of biodegradable cellulose material, and contain sensor modules. Once deployed from a larger aircraft, the Polyplane drone steers itself using tabs attached to the back of each wing. An on-board control system bends each tab to direct the craft as close as possible to a pre-determined landing area. Because the circuits can be ink-jet printed directly into the paper-like material, key components can be glued, instead of being soldered, onto the lightweight, foldable circuit board.

The Samara, designed like a maple seed, has an antenna-equipped circuit board attached to the 'seed' part of the helicopter blade. It cannot be steered, but the design with flexible wings allows the sensor module to float gently to the ground, thus leaving the electronics unharmed and eliminating the need for expensive landing systems.

There are several do-it-yourself drone kits available today, enabling students and hobbyists to develop unmanned aerial vehicles for various non-war purposes—you could try your own too!

Web: <http://robotics.itee.uq.edu.au/wiki/pmwiki.php?n=Site.Research>



(Courtesy: University of Queensland: www.gizmag.com/disposable-uav-paper-plane-maple-seed/26323/pictures#2)

A smart scooter you would love to drive

There is so much talk about smart, connected cars and how they make driving safe and pleasurable. Try extending the concept to scooters, and it might actually be equally beneficial. The recently launched A4000i from Japanese scooter maker Terra Motors is a smartphone-connected e-scooter with a removable battery. The company claims it to be the world's first wireless smartphone-connected e-scooter. An iPhone fits snugly into a compartment designed on the handlebar, and turns into the scooter's dashboard, displaying and synchronising useful information with a Web-based app.

What's inside: The A4000i enables you to fit your iPhone 3G/3GS/4/4S/5 into a dashboard compartment in the front. The e-scooter then synchronises with the iPhone, which starts functioning as the scooter's dashboard. The phone is now capable of displaying mileage, speed and energy use. The same information can also be monitored and analysed on the Web. The application is being updated constantly, and will soon include new features, such as GPS functions, to provide the rider with mapping information or customised advertisements based on their location. Additionally, traffic information could be sent to the Cloud to allow traffic authorities to assess congestion issues and help with urban planning.

Another notable feature of the phone is its specially-designed 48V, 40Ah lithium-based removable battery, which weighs 16 kg and charges in 4.5 hours. The battery lasts for about 50,000 km, providing five times longer battery life than the batteries used in similar electric vehicles.

Web: www.terra-motors.com/scooter/a4000i



The author is a technically-qualified freelance writer, editor and hands-on mom based in Chennai